

Thread pools, or how to share a bounded crew.

From `threading.Thread` to `concurrent.futures` – when a pool helps, and when it just hides the problem.

TOPIC

Thread pools & futures

LANGUAGE

Python 3.12

RUNTIME

75 min read

7.1 Threads & the GIL

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Thread pools

Introduction

CH. 7 · INTRODUCTION

A thread pool is a bounded crew of workers that take jobs from a shared queue. You hand it a function and its arguments; it hands back a **future** — a promise that the answer will be ready later. Pools solve two problems at once: they cap how many threads exist, and they remove the cost of starting a new one for every task.

This chapter assumes you have seen `threading.Thread` but have not yet reached Ch. 8 · `asyncio`. We stay deliberately in the `concurrent.futures` module, which is the right tool for the overwhelming majority of I/O-bound Python programs.¹

WHAT YOU WILL LEARN

- Pool** A fixed-size set of worker threads that consume a work queue.
- Future** An object representing a computation that may not yet have completed.
- Executor** The object you submit work to; it owns the pool and the queue.
- GIL** The Global Interpreter Lock; only one thread runs Python bytecode at a time.

A NOTE ON SCOPE

We cover `thread` pools specifically. Process pools (`ProcessPoolExecutor`) are touched on in §7.4 only to contrast sizing rules. `Async` pools are deferred to chapter 8.

7.1 Threads & the GIL

7.1 · THREADS & THE GIL

Why a pool, and why bounded.

Spawning a thread in Python is cheap but not free. Each thread carries an OS-level stack (8 MB on Linux by default), a bookkeeping structure in the interpreter, and contention for the **GIL**. Unbounded spawning is the single most common cause of a Python server going sideways under load.

7.1.1 Three reasons to pool

1. **Bound memory.** A fixed worker count caps stack usage to a predictable multiple of the stack size.
2. **Amortize startup.** Thread creation is 100 µs; reusing a worker for a 1 ms task matters.
3. **Backpressure for free.** When the queue fills, submitters block — the pool refuses to paper over a too-slow consumer.

7.1.2 When threads do not help

- Pure CPU work in pure Python — the GIL serializes it.
 - Use `ProcessPoolExecutor` instead.
 - Or drop into C via NumPy / `numba`.

The GIL in 3.13. PEP 703 introduces a no-GIL build. Until it is the default, reason as if the GIL is there.

Stack size. Tunable via `threading.stack_size()` — rarely worth doing.

¹For CPU-bound work, substitute `ProcessPoolExecutor`; the API is identical, the cost model is not.

- Work that already releases the GIL (e.g. a `requests` call) benefits regardless.
- Work that calls into a C extension holding a lock of its own.

A thread pool is a queue in a trench coat. Everything interesting is in how the queue behaves when it is full, and in what the workers do when the queue is empty.

7.1.3 Rules of thumb

- ☒ Is the workload I/O-bound?
- ☒ Am I willing to cap memory with a worker count?
- ☐ Can I tolerate out-of-order completion?

7.2 What a pool actually is

7.2 · WHAT A POOL ACTUALLY IS

(This section is listed in the chapter TOC but its content lives in the companion reference card; see Appendix A.)

7.3 Submitting & collecting work

7.3 · SUBMITTING & COLLECTING WORK

The minimal executor.

The standard library ships `concurrent.futures.ThreadPoolExecutor`. It is a context manager; entering it spins the workers, exiting it joins them.

fetch_all.py

PYTHON 3.12

```
from concurrent.futures import ThreadPoolExecutor,
as_completed
import requests

urls = ["https://example.com/", "https://python.org/"]

def fetch(url: str) → int:
    return requests.get(url, timeout=5).status_code

with ThreadPoolExecutor(max_workers=8) as ex:
    futures = {ex.submit(fetch, u): u for u in urls}
    for f in as_completed(futures):
        print(futures[f], f.result())
```

Context manager. Exiting the `with` block calls `shutdown(wait=True)`.

NOTE

Use `as_completed` when order does not matter — results arrive in finish order, not submit order. For submit order, iterate `ex.map` instead.

TIP

Wrap `f.result()` in `try/except`. Exceptions raised inside a worker are deferred until you ask for the result.

WARNING

Never `submit()` from inside a worker of the same pool — you can deadlock. Use a separate pool, or just call the function directly.

DANGER

Do not share an unsynchronised mutable (a `list`, `dict`) across workers. The GIL protects bytecode, not your invariants.

7.4 Sizing the pool

7.4 · SIZING THE POOL

How many workers?



Fig. 1. Submitters push callables into a FIFO work queue; a fixed set of workers pull from it.

A reasonable default for I/O-bound work is given by Little's law:

$$N = \lambda \cdot W \quad (1)$$

where N is the pool size, λ the arrival rate of requests, and W the average time a worker spends per request (mostly blocked on I/O).

Little's law. Kleinrock 1975; applies to any stable queueing system.

WORKLOAD	GOOD DEFAULT	CEILING	WHY
Local disk I/O	4 – 8	32	Kernel queue depth
HTTP calls	8 – 32	256	Remote capacity
DNS lookups	16	64	Resolver cache
Pure Python CPU	1	1	GIL

7.5 Exercises

7.5 · EXERCISES

Work these before 7.6.

Model solutions are in Appendix C, pp. 342–346.

01

SUBMIT / RESULT

Warm-up. Using `ThreadPoolExecutor`, compute the length of ten URLs in parallel and print them in submission order, not completion order.

02

LITTLE'S LAW

Sizing. A service receives 40 requests/second; each spends 0.6 s blocked on a downstream API. Compute the pool size from equation Eq. 1. Justify rounding up.

03

DEADLOCK

Trap. Construct a program where a worker in pool A submits to the same pool and waits on the result. Show the deadlock; fix it with two pools.

Concurrency is a way of organising code. Parallelism is a property of how it runs.

— *Rob Pike* · Concurrency is not parallelism (2012)

7.6 Further reading

- **Kleinrock, L.** Queueing Systems, Vol. 1. Wiley, 1975.
- **Beazley, D.** Python Concurrency From the Ground Up. PyCon US, 2015.
- **CPython source** — `Lib/concurrent/futures/thread.py` .

See also. Ch. 8 · `asyncio`;
Ch. 9 · process pools.

Notes